D7 Defence Brief — Complete System, Readiness and Prepared Answers

Olga Morgunova · 420-977-VA AI Development Project (Capstone 1) · Learning Integration Assessment · release `v1.0` · instructor: Steve Ataky, Ph.D. · evidence collected 2026-11-01 22:59 UTC

This brief is the panel's map of the system and my preparation for the defence. The fifteen-minute talk follows the D6 deck, demonstration first. Every number below is rendered from `reports/metrics/*.json` by `python -m boreal_marche.reporting.build D7`. The readiness table comes from `scripts/collect_d7_evidence.py`, run against the live service.

1. The system the panel can use

Component	Where	State at collection
Interface (EN/FR)	https://boreal-marche-capstone-4ay3m2fn6qdo4jtgy72vpr.streamlit.app	sleeps when unused; the first visit wakes it in ~30 s
API and its documentation	https://padp32qm6p.us-east-1.awsapprunner.com/docs	API `1.0.0`, commit `b00c936` (head of `master`: `b00c936`)
Monitoring dashboard	interface → *Monitoring* tab; `GET /monitoring/report`, `/metrics`	in-process checks every 60 s; hourly external check (`monitor.yml`)
Source, release	`github.com/OlgaMorgunova/boreal-marche-capstone`, tag `v1.0`	tags `v0.1`, `v0.2`, `v0.3`, `v0.9`, `v0.9.5`, `v0.9.6`
Tests	`make cov`	96 passed, 81% line coverage
Recorded defence presentation	https://youtu.be/STCDsmDr3s8	"Boréal Marché Defense" (YouTube)
Fallback video	https://youtu.be/C_Qv7lu3cpY (walkthrough) · https://youtu.be/2bcNpRwMQxE (D6 talk)	in case the network fails

Demonstration path (≈ 4 minutes, all live). 1. *Prediction tab*: first customer → probability, decision, principal factors. Drag "spend in last 180 days" to show the decision move. 2. Open *Campaign list for approval*: coupon / holdout / review / none for 300 customers, and the high-value review list. 3. *Basket tab*: add `22916`, `22918` → herb-marker rules (lift ≈ 140); an unknown item → the popularity fallback, labelled as such. 4. *Assistant tab*: "Combien de temps pour un remboursement ?" → cited French answer; "show me customer BM-…" for someone else → refused; a poem request → out of scope. 5. *Monitoring tab*, then `/docs` → `GET /version`: the commit equals the head of `master`.

2. Readiness check (course Appendix A)

Area	Check	Status	Evidence
live	/health	✓ pass	200
live	/ready	✓ pass	[]
live	/version = head of master	✓ pass	live b00c936fb470 · master b00c936fb470 · API 1.0.0
head A	golden prediction reproduced	✓ pass	p_return=0.406059
head A	campaign action on /predict	✓ pass	coupon
head A	/campaign list (review + hold-out)	✓ pass	{'coupon': 176, 'holdout': 26, 'review': 41, 'none': 57}
head B	/recommend fires a rule	✓ pass	rules

Area	Check	Status	Evidence
head C	/chat answers in French with a citation	✓ pass	tool=policy_search lang=fr citations=4
head C	/chat refuses out-of-scope	✓ pass	out_of_scope
monitoring	dashboard has traffic	✓ pass	5 predictions in window
live	interface reachable (303 = waking up)	✓ pass	303
repository	public	✓ pass	public
repository	ci.yml green on master	✓ pass	success @ b00c936
repository	cd.yml green on master	✓ pass	success @ b00c936
repository	annotated tag v0.1	✓ pass	tag
repository	annotated tag v0.2	✓ pass	tag
repository	annotated tag v0.3	✓ pass	tag
repository	annotated tag v0.9	✓ pass	tag
repository	annotated tag v1.0	… pending	missing
repository	§7.2 structure	✓ pass	22 paths
repository	raw data not committed	✓ pass	only the .dvc pointer
repository	README video above the fold	✓ pass	link in the first 20 lines
cloud	SageMaker spot quota	… pending	CASE_OPENED (case 179073448000753, 4 instances)
manual	Application URL opens in an incognito window on a phone	☐ manual	
manual	Compose stack tested from a fresh clone on a different machine	☐ manual	
manual	Application warm and open in a browser tab before the defence	☐ manual	
manual	Fallback video at hand in case the network fails	☐ manual	
manual	Prepared: primary metric, baseline and threshold with their justification	☐ manual	
manual	Prepared: one answer to 'what would you do differently' and one negative result	☐ manual	

Table 1 — 21 automated checks pass, 0 fail, 2 are pending and 6 need a person. Re-run the morning of the defence; a failure is fixed before the defence, not explained during it.

3. What v1.0 adds: the fairness mitigation is now served

D5 found that one value-blind threshold proposes only 46% of lapsing top-quartile spenders against 89% overall. It adopted, without building, a human review list and a randomised hold-out. Both now run in the API (`retention/campaign.py`). Every `/predict` returns an `action`. `GET /campaign` returns the list the reviewer approves. The automatic proposal is unchanged, so the golden predictions still hold.

- **review**: spend Q3 or Q4, not proposed, but under the M2 group threshold (Q3 ≤ 0.73, Q4 ≤ 0.88). Spend routes the case to a person; it never changes the automatic decision.
- **holdout**: 10% of proposed customers, drawn by `sha256(campaign_id:customer_ref)`. The draw is stateless, auditable and blind to every feature, and a new `campaign_id` each quarter draws a new sample. The customer notice now says so.

Snapshot (served model)	Validation	Test
Proposed / review list / hold-out	1,814 / 368 / 186	1,212 / 450 / 131
Q4 lapser recall: automatic → if the whole list is approved	0.46 → **0.89**	0.16 → **0.82**
Q3 lapser recall: automatic → if approved	0.70 → 0.89	0.34 → 0.66
Lapsers in the review list	26%	28%
Profit if the whole list is approved (assumed save rate 20%)	−$8,221	−$16,921
Hold-out balance: mean `p_return` held out vs treated	0.36 vs 0.35	0.45 vs 0.45
Assumed profit given up by the hold-out (campaign net)	$2,663 ($7,695)	$452 (−$1,039)
Smallest detectable save rate after one quarter (5%, 80% power)	0.16	0.21

Table 2 — `reports/metrics/campaign_policy.json`. These counts reproduce D5's offline audit exactly (M0 and M2 coupons), which checks the served code against the analysis. Campaign profit here uses the served calibrator; D3's $7,421 used out-of-fold calibration.

Read honestly: the list closes the gap only if a person approves it, and approving it all *loses* money under the assumptions. The list is for judgment, not rubber-stamping. The hold-out is the price of finally measuring the save rate: it needs one to two quarters to confirm the assumed 20%.

4. Prepared answers to the question bank (Appendix C)

Data and modelling

4. Windows; why 90 days. Observation 180 days before a cut-off *T*, label = any order in [T, T+90 d). Cut-offs: train 2010-06-01, 2010-09-01, 2010-12-01, 2011-03-01, validation 2011-06-01, test 2011-09-01. Ninety days matches the client's quarterly campaign, and the median gap between orders is about two months, so 90 days separates "slow" from "gone".

5. Why PR-AUC. Classes are near balance overall but the action targets one side. Accuracy hides which errors we make, and ROC-AUC rewards ranking the many easy negatives. PR-AUC measures precision and recall across thresholds. Test 0.847 [0.832, 0.863] against 0.7123 for recency deciles, 0.607 for the prior. I also report the lapse direction (0.6438).

6. Threshold 0.61. It maximises campaign profit on validation under stated costs: $4.00 per coupon, a 15% discount, 20% save rate, 30% margin and a 365-day value horizon. At a 10% save rate the optimum is "contact nobody" (sensitivity table, D6 §5.3). The number is a business decision with its assumptions printed.

7. What calibration buys. We do not only rank. Expected profit multiplies a *probability* by a value, the threshold is in probability units, and monitoring compares mean predicted `p_return` with the realised return rate. Test ECE 0.041.

8. Most important feature. `active_months_180d` (months with an order of the last six), with 50% of attribution. It surprised me that regularity beats recency; recency is partly redundant with it. Top five: `active_months_180d`, `mean_gap_days`, `revenue_180d`, `std_gap_days`, `tenure_days`.

9. If the model loses to the business rule on recall. Measured: "no order in 60 days" as a coupon policy reaches lapser recall 0.56 / 0.61 (val / test) with 1,142 / 1,288 coupons and $2,118 / −$12,182 profit. The model reaches 0.89 / 0.68 and loses less. If it lost on recall I would compare at *equal coupons*: recall alone rises by contacting more people. If it still lost, I would ship the rule and report the model as a negative result. The promotion gate already requires +0.05 PR-AUC over the best baseline.

10. Test protection. One quarter after validation, never used for any choice. Scored 3 times, all disclosed: look 1 before the duplicate-key fix (0.8504), look 2 after the re-run (identical), look 3 for the minimised 19-feature model (0.847). `make evaluate` refuses a new look without `--force`, which records it.

Recommendation

11. Association rules, not collaborative filtering. 22.6% of lines have no customer, baskets are the unit, the catalogue is 4,466 items at 0.54% density, and a rule is explainable ("bought with"). I would switch to implicit-feedback CF or a two-tower model once most traffic is signed in with a history and cross-session personalisation is the goal.

12. The lift trap. `22294 HEART FILIGREE DOVE SMALL` → `85123A WHITE HANGING HEART T-LIGHT HOLDER`: confidence 44.7%, lift 2.998. The consequent is in 14.9% of all baskets, so the rule mostly restates popularity. `min_lift` 3 discards it; the median served lift is 7.9.

13. Why not mine per request. Apriori over 32,126 baskets takes 2.1 s (FP-Growth in mlxtend 241.4 s) and needs the whole history. Mining is offline; serving is an index lookup at 2.2 ms p95.

14. 500,000 rules. Measured with 480,768 rules (64× ours): `/recommend` p95 46 ms ms, 219 MB memory, because the engine indexes rules by antecedent item and checks only matching ones. Beyond that I would prune by lift and support or move rules to a key-value store.

15. No rule matches. The strategy becomes `popularity_fallback`, and the response marks each item `source: popularity` and the interface labels it "Popular with our customers", never "bought with". Coverage: 91.7% of holdout baskets fire a rule; hit-rate@5 0.145 against 0.017 for best-sellers (8.6×).

Generative AI

16. RAG, not fine-tuning. Six policies (4,995 words, 66 chunks) that change. RAG cites the exact chunk and updates by re-indexing. Fine-tuning would bake in stale policies, cannot cite, and needs data and GPUs we do not have.

17. No invented SKUs or probabilities. Product codes and percentages come only from tool output (`/recommend`, `/predict`). An LLM answer containing a SKU or a percentage absent from that output is replaced by the deterministic template answer (`guardrails.py`, rule 5), and a test covers it.

18. Is the assistant good? On 40 golden and 20 paraphrased questions, tool choice is 97.5% / 95% with local `llama3.1:8b`, and out-of-scope refusal is 100%. The keyword-rule provider scores 100% / **35%**, which is why an LLM earns its place. The live service runs the rule provider (no GPU on App Runner): stated, not hidden.

19. Retrieval returns nothing relevant. Below a cosine similarity of 0.30 (chosen by a floor sweep on the golden set), a policy question is handed to a human instead of answered. Tested with out-of-corpus questions in the golden set.

20. Why a local model. Law 25 and the course rule: no personal data to a hosted provider. The local model is the data-residency answer. It also costs $0.23 per 1,000 queries against an estimated $0.19 hosted (unverified price), at 1.6 s per call.

Engineering and MLOps

21. joblib or pickle. joblib is pickle with efficient NumPy arrays. Both execute code on load, so I load only artifacts this pipeline built, pinned to the library versions in `model_meta.json`. XGBoost's JSON or ONNX would remove the risk for untrusted sources.

22. A "model version". The pipeline (preprocessing, estimator, calibrator), the threshold and the feature list in one artifact, plus `model_meta.json`: git commit, data SHA-256, training rows, library versions, MLflow run, registry version. Served now: `0.9.5+xgboost`.

23. /health vs /ready. Health says the process is alive (restart if not). Ready says every artifact is loaded and warm (503 until then). A load balancer sends traffic only to *ready* instances, so a deploy never serves half-loaded containers.

24. OIDC, not keys. One-hour credentials, a trust policy limited to this repository's immutable IDs and the `production` environment, and nothing to rotate or leak. A stored key works from anywhere until someone notices.

25. Retraining opens a pull request. A retrain is a code-and-data change. CI, the promotion gates (baseline, calibration, champion, fairness, minimisation) and a human review apply before deploy. Unattended promotion is how a silent regression ships.

26. Degradation overnight. Input PSI/KS, prediction drift, proposed-coupon rate, errors and p95 are checked every 60 s, alerts fire on level changes, and `monitor.yml` opens a GitHub issue hourly if the status is red or the service is down.

27. A label 90 days late. Meanwhile I monitor inputs, the prediction distribution, coupon rate and data quality. A weekly backfill joins predictions to later orders. The replay showed PR-AUC 0.768 on 2,849 backfilled predictions.

28. When managed training is worth it. When tuning outgrows a laptop: hours not minutes, large data or GPUs. Here XGBoost tuning took 1.6 min on 12,277 rows. The SageMaker job is implemented and dry-run tested. Quota

request: CASE_OPENED, case 179073448000753. Result: not run: the quota request is still open, so every model was tuned locally.

Governance

29. Automated decisions. The coupon proposal (Law 25 s. 12.1). By design it is not exclusively automated: a manager approves every list. We still disclose, give the principal factors on request (`?explain=3`), allow observations to a person, and now also disclose the random hold-out.

30. The recall gap. The options are M1 expected value (profitable, abandons most lapsers), M2 group thresholds (closes the gap, +20% coupons, treats by spend) and M3 re-weighting (halves it, fragile). Chosen: keep one threshold for the automatic decision and send M2's extra cases to a human. Built in v1.0 (§3); the hold-out decides the rest.

31. A hosted LLM outside Canada. Not with customer data until a privacy impact assessment and contractual protection exist (s. 17). Pseudonymise and minimise first; the local provider is the default. Our own serving in `us-east-1` is the same issue, acceptable only because the data is public and pseudonymised; real data means ECS in `ca-central-1`.

32. "Why didn't I get an offer?" Within 30 days (s. 32, to verify): the decision and the main factors from `?explain=3`. If the customer was in the random hold-out, we say so; it is not about them. They can submit observations to a staff member who can review the decision.

5. Negative results to volunteer, and the open limits

- Tuning added < 0.005 PR-AUC; the limit is the features, not the algorithm.
- The first profit model made "contact nobody" optimal until the value horizon was fixed.
- The model does not beat "contact nobody" on the test quarter (−$1,039). It beats the blanket coupon by $54,137.
- The fairness gap is routed to people, not solved; approving the whole review list loses money under the assumptions (§3).
- In mlxtend, Apriori is 116× faster than FP-Growth at this size.
- Open: save rate unmeasured (the hold-out now exists); serving outside Canada; the live log is per-instance SQLite; the hosted LLM provider is unevaluated; the SageMaker run waits on the quota; the LégisQuébec article numbers are to be checked by hand.